

AI Portfolio Optimizer — Technical Report

Project Summary

This system builds an AI-driven portfolio optimizer using real market data from Yahoo Finance. It evaluates asset performance, applies ESG screening, and constructs an optimal portfolio using standard mean-variance optimization.

Data Pipeline

- Daily close prices downloaded for global equities and ETFs.
- Missing values cleaned through forward/backward fill.
- Adjusted Close replaced with Close due to yfinance auto-adjust changes.

Methodology

1. Compute daily returns.
2. Estimate expected annualized returns.
3. Build annualized covariance matrix.
4. Apply ESG filter threshold ≥ 70 .
5. Run Markowitz optimization to maximize Sharpe ratio.

Key Results

- Expected Return: 0.1%
- Expected Volatility: 1.53%
- Sharpe Ratio: 0.0645
- ESG Filter Applied: Yes (≥ 70)

Insights

The optimized portfolio is conservative, driven by stable assets that pass the ESG threshold. ESG reduction narrows diversification but supports sustainability-aligned investing.

Future Enhancements

- Add Bayesian estimation or Black-Litterman modeling.
- Extend to crypto, bonds and commodities.
- Add sentiment scoring from financial news.
- Deploy as real-time dashboard with auto-refresh.
- Integrate scenario stress testing.

